Machine Learning for Kira IBP Reduction

Seed selection and elimination ordering

October 1, 2026

Abstract

Feynman integral calculations are an important component of multiloop amplitude calculations in quantum field theory. In Quantum Chromodynamics (QCD), multiloop amplitudes and their associated Feynman integrals arise in higher-order perturbative calculations and can become increasingly complex at higher loop orders. Recent progress has automated these calculations to some extent using the framework Kira, which organizes target integrals into integral families and sectors, generate algebraic relations among the integrals by using integration by parts (IBP) identities, and reduce the resulting system to expressions in terms of a finite set of master integrals. We aim to explore the application of System 1 machine learning models to accelerate the IBP reduction process by investigating if lightweight learned models can improve seed selection. These models can predict the most useful integrals from which to generate IBP relations while minimizing redundant calculations. We will also explore System 1 models for learning efficient reduction and elimination sequences to minimize the number of algebraic operations required to express target integrals in terms of master integrals. Because the machine learning models will serve as decision-makers while the underlying IBP reduction and verification remain exact, these models can improve computational efficiency without compromising the correctness of the resulting reductions.

1 Introduction

Feynman integrals arise in higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes. In QCD, these corrections involve quark and gluon loops and contribute to predictions for cross sections and decay rates. Calculating a loop contribution requires integration over the independent internal momenta that remain after imposing momentum conservation [1].

The scalar integrals that appear in an amplitude have the general form

$$\mathcal{J} = \left(\prod_{k=1}^{L} \int \frac{\mathrm{d}^d \ell_k}{\mathrm{i}\pi^{d/2}}\right) \frac{\mathcal{N}(\boldsymbol{\ell}, \boldsymbol{p})}{\prod_{i=1}^{n_{\mathrm{phys}}} D_i^{a_i}}, \qquad D_i = q_i^2 - m_i^2 + \mathrm{i}0. \tag{1}$$

Here L is the number of loops and $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_L)$ is the list of independent loop momenta over which we integrate. The number of independent external momenta is E, and we keep these momenta, $\boldsymbol{p} = (p_1, \ldots, p_E)$, fixed during integration. The numerator $\mathcal{N}$ is a polynomial obtained from the spin and tensor algebra; we leave its mass dependence implicit in the notation. The number of physical denominators is n_{phys}, and each D_i corresponds to an internal particle line with momentum q_i and mass m_i. The positive integer a_i is the power of D_i in the denominator, and q_i is a fixed linear combination of loop and external momenta, with $q_i^2 = q_i \cdot q_i$ denoting the Lorentz scalar product.

The integration dimension is $d = 4 - 2\epsilon$, where ϵ is the regulator used to define divergent integrals, so the result depends on ϵ as well as the external momenta and masses. The factor $1/(\mathrm{i}\pi^{d/2})$ is our normalization convention for each integration measure, with $\mathrm{i}^2 = -1$. In each denominator, i0 is an infinitesimal positive imaginary term that specifies how to integrate past a propagator pole, where $q_i^2 = m_i^2$.

Higher loop orders produce collections of integrals with related numerators and denominators and algebraic identities can be used to express them in terms of an independent set of master integrals. Kira automates this reduction [2], after which the calculation requires the values of the master integrals. To explain how the reduction works and where machine learning (ML) could speed it up, we first define the notation used in this paper, then derive the identities and describe how Kira solves the resulting equations.

2 The mathematical form of Feynman integrals

To label the integrals related to Eq. (1), we specify their momentum dependence and write the numerator and denominator powers as a vector of integers. This vector identifies each integral as an unknown in the reduction equations.

2.1 Loop momenta, propagators and regularization

After choosing how the loop and external momenta flow through the internal lines, we write each physical denominator as

$$D_i = q_i^2 - m_i^2 + \mathrm{i}0, \qquad q_i = \sum_{k=1}^{L} \alpha_{ik} \ell_k + \sum_{\beta=1}^{E} \eta_{i\beta} p_\beta. \tag{2}$$

The constants α_{ik} and $\eta_{i\beta}$ specify which loop and external momenta make up q_i, with indices k, j for loop momenta and β, γ for external momenta. Squaring q_i gives the scalar products $\ell_k \cdot \ell_j$, $\ell_k \cdot p_\beta$ and $p_\beta \cdot p_\gamma$, and those containing loop momenta vary during integration. The

vector $\boldsymbol{x}$ contains the independent external scalar products and squared masses, which are the kinematic parameters we keep fixed during integration.

Loop integrals can diverge at large momenta or when massless momenta become small or parallel. Dimensional regularization defines these integrals at values of d for which they converge and extends the result by analytic continuation to $d = 4 - 2\epsilon$, using additional regulators where necessary. In an expansion around $\epsilon = 0$, the divergences appear as negative powers of ϵ, which renormalization and infrared cancellation in the observable address later in the calculation [1]. Having specified the regulated measure and momentum dependence, we can now use the powers of the numerator and denominator factors to label individual integrals.

2.2 A common notation for an integral family

We add auxiliary expressions to the physical denominators so that we can write every required loop-momentum scalar product in terms of the list $D_1, \ldots, D_N$ and the fixed parameters $\boldsymbol{x}$. Here N is the number of expressions in the list, and the auxiliary expressions allow us to represent numerator factors. Writing the numerator in terms of this list expresses $\mathcal{J}$ as a linear combination of integrals of the form

$$I_{\boldsymbol{a}}(\boldsymbol{x}, \epsilon) = \int_\ell \prod_{i=1}^{N} D_i^{-a_i}, \qquad \int_\ell \equiv \prod_{k=1}^{L} \int \frac{\mathrm{d}^d \ell_k}{\mathrm{i}\pi^{d/2}}, \qquad \boldsymbol{a} = (a_1, \ldots, a_N) \in \mathbb{Z}^N. \tag{3}$$

Here $\int_\ell$ is shorthand for the loop integration measure and $\boldsymbol{a}$ is the vector of N integer powers, with $\mathbb{Z}$ denoting the set of integers. These powers can be positive, zero or negative, so we can write numerator and denominator factors in the equivalent form

$$I_{\boldsymbol{a}}(\boldsymbol{x}, \epsilon) = \int_\ell \frac{\prod_{i:a_i<0} D_i^{-a_i}}{\prod_{i:a_i>0} D_i^{a_i}}. \tag{4}$$

A positive index places $D_i^{a_i}$ in the denominator, a negative index places $D_i^{-a_i}$ in the numerator, and a zero index removes the factor, with an empty product equal to one. Auxiliary expressions used only as numerators therefore have nonpositive indices. An irreducible scalar product is a numerator scalar product that cannot be written as a linear combination of the physical denominators and terms depending only on $\boldsymbol{x}$ [3].

An integral family contains the integrals obtained by varying $\boldsymbol{a}$ for a fixed list $D_1, \ldots, D_N$ and fixed kinematics. Within this family, the target list specifies the integrals that the amplitude requires, and grouping family members by their positive indices organizes them into sectors.

2.3 Sectors and the organization of related integrals

A sector is a set of integrals with the same positive-index positions, so they contain the same denominators. We assign a label to each sector using

$$\theta_i(\boldsymbol{a}) = \begin{cases} 1, & a_i > 0, \\ 0, & a_i \leq 0, \end{cases} \qquad S(\boldsymbol{a}) = \sum_{i=1}^{N} 2^{i-1} \theta_i(\boldsymbol{a}). \tag{5}$$

The indicator θ_i is one when denominator i is present and zero otherwise, and S is an integer that identifies this set of denominators. Integrals with different powers of the same denominators belong to the same sector because S depends only on which indices are positive. Removing one or more denominators gives a subsector containing a proper subset of the original sector's denominators.

Scaleless sectors contain no nonzero mass or kinematic scale and therefore vanish in dimensional regularization, while changes of loop integration variables can yield symmetry relations between integrals. Kira uses these zero sectors and symmetry mappings to reduce the number of unknowns [2]. We can then use the sector and index labels to write the identities that relate the remaining family members.

3 From integration to algebraic reduction

To express the targets in terms of master integrals, we generate relations between family members through integration by parts and solve the resulting equations. Evaluating the master integrals then gives the target integral values needed to calculate the amplitude.

3.1 How integration by parts produces relations

For differentiable functions $f(z)$ and $g(z)$ of a real variable z, the product rule gives the integration identity between endpoints z_0, z_1:

$$\int_{z_0}^{z_1} f'(z)g(z)\,\mathrm{d}z = [f(z)g(z)]_{z_0}^{z_1} - \int_{z_0}^{z_1} f(z)g'(z)\,\mathrm{d}z. \tag{6}$$

Primes denote derivatives with respect to z, and the brackets denote the endpoint difference $f(z_1)g(z_1) - f(z_0)g(z_0)$. To apply this product rule to loop-momentum derivatives, we derive the integration-by-parts (IBP) identity where the regulated integral has a vanishing boundary term and then extend it by analytic continuation [4, 3]. For a loop momentum ℓ_k and a vector $v \in \{\ell_1, \ldots, \ell_L, p_1, \ldots, p_E\}$, the identity is

$$0 = \int_\ell \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_{i=1}^{N} D_i^{-a_i} \right). \tag{7}$$

The index μ is a spacetime index, and a repeated Lorentz index means that we sum over components using the spacetime metric. The derivative acts on both v^μ and the factors $D_i^{-a_i}$, so the product rule gives

$$0 = \int_\ell \left[\frac{\partial v^\mu}{\partial \ell_k^\mu} - \sum_{i=1}^{N} a_i \frac{v^\mu (\partial D_i / \partial \ell_k^\mu)}{D_i} \right] \prod_{j=1}^{N} D_j^{-a_j}. \tag{8}$$

The first term equals d for $v = \ell_k$ and vanishes when v is an external momentum or another independent loop momentum. Differentiating an inverse propagator produces scalar products, which we express through the family basis to convert each term into an integral with shifted indices:

$$\sum_{j=1}^{n_\lambda} c_{\lambda j}(\boldsymbol{a}; \boldsymbol{x}, \epsilon) I_{\boldsymbol{a}+\boldsymbol{\Delta}_{\lambda j}}(\boldsymbol{x}, \epsilon) = 0. \tag{9}$$

Here λ identifies the chosen pair (k, v), n_λ is the number of terms, and $\boldsymbol{\Delta}_{\lambda j}$ is the integer change in the index vector for term j. Differentiation gives the coefficient $c_{\lambda j}$, which depends on the integral indices and the parameters $\boldsymbol{x}, \epsilon$ [4].

The L choices of differentiated momentum and $L+E$ choices of v give $L(L+E)$ basic identities at each index vector, some of which may depend linearly on the others. Further relations follow from Lorentz invariance because a common rotation or Lorentz boost of all external momenta preserves a scalar integral, giving the infinitesimal identity

$$\sum_{\beta=1}^{E} \left(p_\beta^\nu \frac{\partial}{\partial p_{\beta\mu}} - p_\beta^\mu \frac{\partial}{\partial p_{\beta\nu}} \right) I_{\boldsymbol{a}} = 0. \tag{10}$$

Here μ, ν are free spacetime indices, meaning that the identity holds for each pair of components, and the derivatives act on the external-momentum dependence of the integral. Contracting these indices with external momenta gives scalar relations [5] that join the IBP and symmetry identities to form the reduction system.

3.2 From relations to master integrals

Let n_I be the number of distinct integrals in the generated equations and n_{eq} the number of equations. We arrange these integrals in a column vector so that the system takes the matrix form

$$A(\boldsymbol{x}, \epsilon)\, \boldsymbol{I} = 0, \qquad \boldsymbol{I} = (I_{\boldsymbol{a}^{(1)}}, \ldots, I_{\boldsymbol{a}^{(n_I)}})^{\mathsf{T}}. \tag{11}$$

The superscript T denotes transpose, so $\boldsymbol{I}$ is a column vector of integrals whose dependence on $\boldsymbol{x}, \epsilon$ we leave implicit here. The matrix A has n_{eq} rows and n_I columns, with each entry A_{ij} giving the coefficient of integral j in equation i. Solving this system expresses dependent integrals in terms of an independent set. This set is a master basis if the reduction relations express every required integral as a linear combination of its members. For the standard class of Feynman integrals at fixed graph, this basis contains a finite number of masters [6]. We write them as $M_1, \ldots, M_{n_M}$, where n_M is the number of master integrals, so the reduction formula is

$$I_{\boldsymbol{a}}(\boldsymbol{x}, \epsilon) = \sum_{m=1}^{n_M} C_{\boldsymbol{a}m}(\boldsymbol{x}, \epsilon) M_m(\boldsymbol{x}, \epsilon). \tag{12}$$

Here $C_{\boldsymbol{a}m}$ is the coefficient of master M_m in the reduction of target $I_{\boldsymbol{a}}$. The solver obtains it by adding, multiplying and dividing the identity coefficients, so it is a rational function, meaning a ratio of polynomials in $\boldsymbol{x}, \epsilon$. Different independent sets can define different master bases, and an insufficient equation set can leave additional integrals unresolved. Establishing the master basis therefore requires checking that the generated equations are enough to reduce the required integrals.

Equation (12) gives the coefficients multiplying the masters, whose evaluation requires a further calculation, for example by solving differential equations with known values or limits as boundary conditions [5]. Kira provides the reduction rules needed for this calculation by generating and processing the identities through the procedure described next.

4 How Kira performs the reduction

Kira implements the Laporta approach by generating a finite set of identities and eliminating integrals according to an ordering [7]. To obtain the coefficients in Eq. (12), Kira selects seeds, removes redundant equations and solves the retained system. We examine how Kira performs these steps to specify the seed and elimination choices that the proposed models would make.

4.1 Kira's input and seed generation

The user supplies the families in Eq. (3), relations among the parameters $\boldsymbol{x}$, the target list, and bounds on equation generation so that Kira can initialize the families and sectors. During this initialization, Kira detects scaleless sectors and identifies symmetry relations and mappings between compatible families [2].

A seed specifies the index vector $\boldsymbol{a}$ at which Kira evaluates Eq. (9) to generate several derivative identities, and the seed set may include integrals beyond the target list. The shifts $\boldsymbol{\Delta}_{\lambda j}$ in these

identities can introduce further integrals outside the seed set, so the resulting system depends on both the chosen seeds and the terms generated from them. To set limits on the indices of the seeds, Kira uses the quantities

$$t(\boldsymbol{a}) = \sum_{i=1}^{N} \theta_i(\boldsymbol{a}), \qquad r(\boldsymbol{a}) = \sum_{a_i>0} a_i, \tag{13}$$

$$s(\boldsymbol{a}) = -\sum_{a_i<0} a_i, \qquad n_{\mathrm{dot}}(\boldsymbol{a}) = r(\boldsymbol{a}) - t(\boldsymbol{a}). \tag{14}$$

Here t is the number of distinct denominators, r is the sum of their powers, and s is the sum of the numerator powers. The quantity n_{dot} is the sum of the extra denominator powers beyond one for each denominator. We use this symbol to keep d reserved for the integration dimension, and we restrict equation generation by specifying the allowed sectors and limits

$$r \le r_{\max}, \qquad s \le s_{\max}, \qquad n_{\mathrm{dot}} \le n_{\mathrm{dot,max}}, \tag{15}$$

where $r_{\max}$, $s_{\max}$ and $n_{\mathrm{dot,max}}$ are the upper limits on the corresponding quantities. The user sets $r_{\max}$ and $s_{\max}$ and may also set $n_{\mathrm{dot,max}}$. Because these choices determine which seeds contribute equations, a reduction run must establish whether the bounds supply enough relations for the targets [8].

Kira 3 changes seeding in symmetry-related sectors and provides the optional `truncate_sp` setting to restrict numerator seeding in subsectors, with parameter choices that require checks against the expected master basis [9]. The generated system can still contain redundant equations, which elimination identifies as it processes the relations into reduction rules.

4.2 Integral ordering, sector order and forward elimination

Elimination solves an identity for one integral and substitutes the result into other identities, using an integral ordering to select which unknown to eliminate from each equation. Kira defines these orderings through sector labels and the powers of the numerator and denominator factors, with choices that prioritize numerator powers or repeated denominators and thereby affect the master basis and runtime [8].

To describe this procedure, write $I_j = I_{\boldsymbol{a}^{(j)}}$ for the integrals in Eq. (11) and choose the ordering $I_1 \succ I_2 \succ \cdots \succ I_{n_I}$, where $\succ$ gives priority for elimination. In a row whose first nonzero term contains I_k, let b_j denote the coefficient of I_j, so choosing the nonzero coefficient b_k as the pivot gives

$$b_k I_k + \sum_{j>k} b_j I_j = 0 \quad \Longrightarrow \quad I_k = -\sum_{j>k} \frac{b_j}{b_k} I_j. \tag{16}$$

The coefficients b_j depend on $\boldsymbol{x}, \epsilon$, and substituting this expression into the other rows removes their dependence on I_k. Repeating this step gives forward elimination, which builds a triangular system in which each pivot integral depends on integrals later in the ordering. Back substitution then replaces the remaining pivot integrals using their own rules, completing the Laporta solution [7].

Sector scheduling sets the order in which the solver processes groups of integrals or equations, taking into account which other integrals their equations contain. Ordinary IBP identities for a sector involve that sector and its subsectors because nonpositive indices remain nonpositive when applying the derivative. Equations for one sector can therefore depend on integrals from its subsectors. The Kira 3 description starts equation generation with sectors earlier in its ordering and inserts previously obtained pivot relations into new equations [9], with the order of processing depending on the stage of the reduction.

When several rows share the same first integral, the choice of pivot row changes the substitutions that follow. Choosing row ρ to eliminate column k, for example, updates row i by

$$A_{ij} \longleftarrow A_{ij} - \frac{A_{ik}}{A_{\rho k}} A_{\rho j}, \qquad A_{\rho k} \neq 0. \tag{17}$$

Here i is the index of a row other than the pivot row, so $i \neq \rho$, and j is the column index. Subtracting the stated multiple of row ρ sets the coefficient in column k to zero and can create nonzero entries in other columns, a change called fill-in when a previously zero entry becomes nonzero. These substitutions also make the coefficient expressions longer by combining their numerators and denominators, so elimination can require more time and memory as it proceeds. Redundant equations add to this work [10], so Kira uses modular elimination to select equations before processing their full symbolic coefficients.

4.3 Removing redundant equations and selecting target dependencies

To reduce the number of equations it must solve, Kira identifies rows that follow from other equations. Elimination reduces every coefficient in such a row to zero, allowing the solver to omit that row. Kira tests these dependencies with modular arithmetic [2] by substituting sample values $\boldsymbol{x}_0, \epsilon_0$ and mapping the coefficients to a finite field $\mathbb{F}_P$ for a prime integer P:

$$A(\boldsymbol{x}, \epsilon) \longmapsto A(\boldsymbol{x}_0, \epsilon_0) \bmod P. \tag{18}$$

Integers from 0 to $P - 1$ represent the elements of $\mathbb{F}_P$, and addition and multiplication keep results in this range by taking the remainder upon division by P. Division uses the modular inverse of a nonzero element, which multiplies that element to one modulo P, so the sampling procedure excludes points where a required denominator vanishes. This arithmetic keeps field elements bounded in size while allowing the solver to identify independent rows and retain their original symbolic identities.

The matrix rank is the largest number of independent rows, where no row is a linear combination of the others in the chosen set. Some sample values can make otherwise independent rows dependent, giving a rank smaller than that of the symbolic matrix. Checks must account for this possibility when using the probabilistic equation-selection method developed by Kant [10].

Kira also tracks which independent equations contribute to the target reductions, using the dependencies of the forward-eliminated system for this selection in Kira 3 [9]. Seed selection therefore supplies the integrals from which Kira generates equations, and equation selection then retains the relations needed for the targets so that the solver can perform the full reduction on this smaller system.

4.4 Back substitution and the final symbolic result

To obtain $C_{\boldsymbol{a}m}$ from the retained equations, Kira applies back substitution to the pivot rules produced by forward elimination. Since these rules can contain other pivot integrals, back substitution inserts their expressions recursively until the right-hand sides contain only masters, giving Eq. (12). Kira's original symbolic solver carries out this calculation using rational-function arithmetic through the computer algebra system Fermat [2].

Kira also solves systems at finite-field sample points and reconstructs the rational coefficients with FireFly, the reconstruction library introduced into the solver in Kira 2.0 [11]. In this procedure, each sampled coefficient has the form

$$C_{\boldsymbol{a}m}(\boldsymbol{x}_0, \epsilon_0) \bmod P. \tag{19}$$

The modular operations use exact arithmetic, and reconstruction recovers $C_{\boldsymbol{a}m}(\boldsymbol{x}, \epsilon)$ from sufficient samples before checking the result at additional points. Reconstruction therefore forms part of the calculation of the symbolic reduction coefficients, after which the master integrals still require evaluation [11].

Because seed selection determines the equations that enter this procedure and elimination decisions determine the substitutions and intermediate expressions, the proposed ML models would guide these two operations to reduce the time and memory needed for the reduction.

5 ML for seed selection and elimination ordering

To reduce the time and memory needed for IBP reduction, we will study two System 1 models: one for seed selection and one for elimination ordering. Here a System 1 model is a small machine learning model that learns from recorded reductions and makes a choice from the information available at a given stage. We would add the seed model before equation generation and the elimination model at the stage where the solver selects a pivot. Each model would assign scores to the available choices, and Kira would perform the algebra for the selected choice. Training these models would require target integrals with known master expansions, the seeds that supply their reduction equations, and the steps used to solve those equations.

5.1 Selecting seeds before equation generation

To select seeds, the first model would receive the family definition, the target index vectors $\boldsymbol{a}$, the chosen master basis and a set of candidate seeds within specified sectors and index bounds. For each candidate, it would use the seed indices and the target information to estimate how useful the resulting equations would be for the requested reduction.

During training, we would pair each target with the seed sets recorded in its known reductions. These examples would teach the model to rank seeds that together provide enough equations to reduce the target. Different reductions can use different sufficient seed sets, so the training data would retain these alternatives and their equation counts. When applying the trained model, we would select highly ranked seeds and ask Kira to generate their identities. If the equations leave a target unresolved in the chosen master basis, we would add further seeds and repeat the reduction. The second model would guide the elimination of the resulting equations.

5.2 Choosing elimination steps within the solver

To limit fill-in and the growth of coefficient expressions, the second model would rank the allowed pivot choices using the current equations, their nonzero coefficients and the integral ordering. For each recorded reduction step, the training example would contain the equations before the choice, the available pivots, the selected row and integral, and the operations that followed. The remaining recorded steps and their operation counts would show how that choice contributed to a complete reduction.

During a reduction, the model would select an allowed pivot and Kira would carry out the row operations in Eq. (17). The model would then receive the updated system and select the next pivot, continuing until Kira can complete the reduction by back substitution. Each choice must respect the integral ordering and chosen master basis. Preparing the model inputs and obtaining its scores must take less time than its choices save in the algebraic calculation. Kira would use its existing reconstruction procedure for calculations over finite fields.

5.3 Integral data required for training and testing

To train and test these models, we require examples that specify the family denominators, kinematic parameters, targets and master basis, together with the exact coefficients in Eq. (12). For a target containing several integrals, these coefficients specify the master expansion of the complete expression. Each example must also identify the source equations, their seeds and derivative operators, and any symmetry relations used, so the seed model's choices determine the equations available to the elimination model.

We have constructed an initial collection with these data and retained the operations used both to form each target from masters and to reduce it back to the same basis. The elimination records specify the target's coefficients before and after each step, the available source rows, the selected pivot and the operation counts. These records describe elimination of a target coefficient row against a fixed set of source identities. Using the model within Kira will also require inputs and allowed choices that match Kira's evolving equation system.

The exact master coefficients supply the reference answer for checking complete reduction in the chosen basis, as discussed in Section 3 [9]. We keep these answers and construction histories separate from the problem information supplied to the models. Section 6 explains how we obtained the master bases, constructed the targets and checked the associated seed and elimination records.

6 Target Integral Generation for Machine Learning

To train and test the two models described in Section 5, we constructed targets from known master bases while retaining the identities, seeds and algebraic steps used in each construction. We first identified 49 master entries across 16 one-loop and two-loop families with Kira 3.1, and then used these bases to construct 1,662 targets with exact master expansions. This collection contains 309 individual integrals and 1,353 expressions containing several integrals [12, 13]. The following subsections define the families, explain the basis checks and describe how the reverse construction supplies the data needed by the models. All integrals remain symbolic throughout this procedure.

6.1 One-loop families and their kinematics

We use the integral definition and normalization in Eq. (3), with $d = 4 - 2\epsilon$ and inverse propagators $D_i = q_i^2 - m_i^2 + \mathrm{i}0$. For each family, we specify the internal momenta q_i, the masses m_i and the scalar products of the external momenta, so every index vector defines a particular integral. In the equations below we suppress the common i0 prescription. We write $\mu = m^2$ for a common nonzero squared mass and $\mu_1 = m_1^2$, $\mu_2 = m_2^2$ for two independent nonzero squared masses. The symbols x_s and x_t denote momentum invariants and remain distinct from the index sums s and t in Eq. (14).

The one-loop collection begins with a massive tadpole, whose single inverse propagator is $D_1 = \ell_1^2 - \mu$ and which has no external momentum. We then include four bubble families with external momentum p and inverse propagators

$$D_1 = \ell_1^2 - m_1^2, \qquad D_2 = (\ell_1 - p)^2 - m_2^2, \qquad p^2 = x_s. \tag{20}$$

The squared-mass pairs are $(0, 0)$, $(\mu, 0)$, (μ, μ) and (μ_1, μ_2), corresponding to a massless bubble, a bubble with one massive line, an equal-mass bubble and an unequal-mass bubble. We keep x_s as an independent nonzero scale, with the massive families defined at generic values of their independent parameters.

To include three-point integrals, we define triangle families by

$$\begin{aligned} &D_1 = \ell_1^2 - m_1^2, \quad D_2 = (\ell_1 - p_1)^2 - m_2^2, \qquad D_3 = (\ell_1 - p_1 - p_2)^2 - m_3^2, \\ &p_1^2 = u_1, \qquad\quad p_2^2 = u_2, \qquad\quad (p_1 + p_2)^2 = u_3. \end{aligned} \tag{21}$$

Here u_1, u_2, u_3 are the squared external momenta, also called external virtualities, and $p_1 \cdot p_2 = (u_3 - u_1 - u_2)/2$ fixes the remaining external scalar product. We use three massless families with $(u_1, u_2, u_3) = (0, 0, x_s)$, $(0, x_2, x_3)$ and (x_1, x_2, x_3). The parameters x_1, x_2, x_3 denote independent nonzero virtualities. We also use two families with $m_1^2 = m_2^2 = m_3^2 = \mu$ and virtualities $(0, 0, x_s)$ and (x_1, x_2, x_3). Thus the five triangle families vary both the internal masses and the number of nonzero external virtualities.

For four-point integrals, we use three massless box families with $p_4 = -p_1 - p_2 - p_3$ and

$$\begin{aligned} &D_1 = \ell_1^2, \qquad\quad D_2 = (\ell_1 - p_1)^2, \\ &D_3 = (\ell_1 - p_1 - p_2)^2, \quad D_4 = (\ell_1 - p_1 - p_2 - p_3)^2, \\ &x_s = (p_1 + p_2)^2, \qquad x_t = (p_2 + p_3)^2. \end{aligned} \tag{22}$$

Writing $u_i = p_i^2$ for $i = 1, \ldots, 4$, we fix the independent external scalar products through

$$p_1 \cdot p_2 = \frac{x_s - u_1 - u_2}{2}, \qquad p_2 \cdot p_3 = \frac{x_t - u_2 - u_3}{2}, \qquad p_1 \cdot p_3 = \frac{u_4 + u_2 - x_s - x_t}{2}. \tag{23}$$

The virtuality choices are $(0, 0, 0, 0)$, $(0, 0, 0, x_4)$ and (x_1, x_2, x_3, x_4), where x_4 is the fourth nonzero virtuality parameter. These choices give boxes with zero, one and four off-shell legs; in this classification an on-shell external leg has $p_i^2 = 0$. We treat the remaining invariants as independent subject to the stated relations. These definitions give 13 one-loop families, which we extend with the following three two-loop families.

6.2 Two-loop families and auxiliary expressions

The two-loop families use loop momenta ℓ_1, ℓ_2. For a massless two-point integral with external momentum p and $p^2 = x_s$, we take

$$(D_1, D_2, D_3, D_4, D_5) = (\ell_1^2, \ell_2^2, (\ell_1 - \ell_2)^2, (\ell_1 - p)^2, (\ell_2 - p)^2). \tag{24}$$

These five expressions determine the five loop-dependent scalar products $\ell_1^2, \ell_2^2, \ell_1 \cdot \ell_2, \ell_1 \cdot p$ and $\ell_2 \cdot p$. Varying their indices therefore includes numerator insertions as well as repeated propagators.

For the equal-mass sunrise with the same external invariant, we use

$$\begin{aligned} &D_1 = \ell_1^2 - \mu, \qquad D_2 = \ell_2^2 - \mu, \qquad D_3 = (p - \ell_1 - \ell_2)^2 - \mu, \\ &D_4 = (p - \ell_1)^2, \quad D_5 = (p - \ell_2)^2, \quad a_4, a_5 \leq 0. \end{aligned} \tag{25}$$

The first three expressions are the physical inverse propagators. The auxiliary expressions D_4, D_5 complete the five scalar products needed to express the derivatives in the IBP identities. Their nonpositive indices restrict them to numerator factors or their absence, so the top sector contains only D_1, D_2, D_3 as denominators. We keep x_s and μ independent.

The final family is the equal-mass two-loop vacuum sunset, which has no external momentum and uses

$$D_1 = \ell_1^2 - \mu, \qquad D_2 = \ell_2^2 - \mu, \qquad D_3 = (\ell_1 - \ell_2)^2 - \mu. \tag{26}$$

These three expressions determine all loop scalar products for this vacuum family. For every family, we include the sector containing all physical propagators and all its subsectors, so the

resulting basis also accounts for integrals with removed propagators. The number of physical propagators in this top sector is denoted by t_{top}. It is one, two, three and four for the tadpole, bubble, triangle and box families, respectively, five for the massless two-loop propagator, and three for the sunrise and vacuum families. This number sets the initial range of identities used to identify the masters.

6.3 Identification and checks of the master bases

For each family, we used Kira 3.1 to generate IBP relations together with the applicable Lorentz-invariance and symmetry relations. Kira removes scaleless sectors and uses finite-field elimination to identify the integrals that remain independent in the equations needed for a selected integral range [2, 9]. We used Kira's integral ordering 1 throughout, which orders sectors by their labels and prefers numerator insertions over extra denominator powers when selecting representatives within a sector [8]. Keeping this ordering fixed gives a specified basis against which to compare later calculations.

We first generated identities from seeds satisfying

$$r \leq t_{\text{top}} + 2, \qquad s \leq 2, \qquad n_{\text{dot}} \leq 2, \tag{27}$$

and selected integrals to probe the basis within

$$r \leq t_{\text{top}} + 1, \qquad s \leq 1, \qquad n_{\text{dot}} \leq 1. \tag{28}$$

Here r is the sum of positive indices, s is minus the sum of negative indices and $n_{\text{dot}} = r - t$ is the number of extra denominator powers, as defined in Eq. (14). Both sets of limits apply to the top sector and its subsectors. The probe integrals specify the range for which Kira must identify the required masters, while the wider seed range supplies relations that can eliminate the additional integrals introduced by index shifts. These probes serve the basis calculation; the ML examples come from the subsequent reverse construction.

Kira evaluated the coefficients of these relations over a finite field using the substitution in Eq. (18). For the first calculation we used the prime $P_1 = 2147483647$ and dimension value $d_1 = 37$. To specify the kinematic substitutions, define the ordered parameter list

$$\boldsymbol{z} = (x_s, x_t, x_1, x_2, x_3, x_4, \mu, \mu_1, \mu_2). \tag{29}$$

The first sample was

$$\boldsymbol{z}^{(1)} = (-13, -19, -11, -17, -29, -43, 7, 7, 23), \tag{30}$$

using only the entries present in each family and retaining all zeros and equal-mass conditions in its definition. We interpret these assignments modulo P_1 when evaluating the identity coefficients. The dimension sample corresponds to $\epsilon_1 = (4 - d_1)/2$ in the same field. These substitutions test algebraic relations; the master labels continue to refer to integrals at symbolic $\boldsymbol{x}, \epsilon$.

We then enlarged the seed and probe ranges to check whether the initial basis remained sufficient:

$$\begin{aligned} &\text{seeds:} \quad r \leq t_{\text{top}} + 3, \qquad s \leq 3, \qquad n_{\text{dot}} \leq 3, \\ &\text{probes:} \quad r \leq t_{\text{top}} + 2, \qquad s \leq 2, \qquad n_{\text{dot}} \leq 2. \end{aligned} \tag{31}$$

We repeated the calculation first at the original sample and then at a second sample with prime $P_2 = 1000000007$, dimension $d_2 = 41$ and

$$\boldsymbol{z}^{(2)} = (-31, -47, -23, -37, -53, -71, 11, 11, 29). \tag{32}$$

Changing the prime, the dimension and the independent kinematic parameters provides a second check of the linear dependencies, since particular substitutions can lower the rank of the symbolic coefficient matrix [10].

In both enlarged calculations, we supplied the original masters as the preferred basis and enabled Kira's check for additional or unused masters [9]. We required the complete list of master index vectors to agree in all three calculations, and also checked each vector against its family definition and sector label. All 16 families satisfied these requirements, giving 48 completed master-identification calculations. The agreement provides evidence for basis stability over the tested index ranges and sample points. Completeness for arbitrary integer indices would require further relations or an independent argument, and special kinematic limits can change the basis.

We stopped each calculation after master identification. This stage determines which symbolic integrals form the starting basis within the checked domains. We then obtained rational reduction coefficients for the constructed targets through the algebraic procedure described below. The numerical values and ϵ expansions of the master integrals remain separate from this basis determination.

6.4 Resulting bases and the starting coefficients

The massive tadpole gave one master. The massless, one-mass, equal-mass and unequal-mass bubbles gave one, two, two and three masters, respectively. The massive bubble bases include the necessary tadpole subsectors, with two distinct tadpoles when the internal masses differ.

The massless triangles with one, two and three nonzero external virtualities gave one, two and four masters, respectively. For the first two families, Kira chose bubble integrals in subsectors as the masters. The three-virtuality family also retained a triangle integral. The equal-mass triangles with one and three nonzero virtualities gave three and five masters. The massless boxes with zero, one and four off-shell legs gave three, four and eleven masters, respectively, including the required lower sectors in each count.

The massless two-loop propagator gave two masters, the equal-mass sunrise gave three, and the equal-mass vacuum sunset gave two. For the denominator order in Eq. (25), the sunrise basis is

$$M_1 = I_{(1,1,0,0,0)}, \qquad M_2 = I_{(1,1,1,0,0)}, \qquad M_3 = I_{(1,1,1,-2,0)}. \tag{33}$$

The first integral factorizes into a product of two massive tadpoles because removing D_3 separates the loop integrations. The other two integrals retain all three sunrise propagators, and M_3 contains the numerator factor D_4^2. This example shows how the chosen ordering can retain a numerator integral as a master.

The counts sum to 49 master entries across the 16 families. We count each basis within its own family, so shared tadpole and bubble subsectors can appear in more than one family. The complete family definitions and master index vectors are available with the accompanying data [12]. Each entry specifies the denominators, kinematics, index vector, sector and mass dimension. With the normalization in Eq. (3), the latter is

$$[I_{\boldsymbol{a}}]_{\text{mass}} = Ld - 2\sum_{i=1}^{N} a_i, \tag{34}$$

where $[I_{\boldsymbol{a}}]_{\text{mass}}$ is the power of mass carried by the integral. Each loop measure contributes d powers and each factor $D_i^{-a_i}$ contributes $-2a_i$. This dimension constrains the coefficients used to combine integrals in the target construction.

For that construction, we kept the identified masters symbolic and initialized each master as itself. If $\boldsymbol{\mu}^{(m)}$ is the index vector of master M_m, the starting coefficients are

$$M_m = I_{\boldsymbol{\mu}^{(m)}}, \qquad C_{\boldsymbol{\mu}^{(m)}n} = \delta_{mn}, \qquad m, n = 1, \ldots, n_M. \tag{35}$$

Here n_M is the number of masters in the selected family, and δ_{mn} is one when $m = n$ and zero otherwise. Thus each master has coefficient one for itself and zero for the others. We held the family definition and basis fixed while constructing its targets, starting with these masters and extending the collection through the identities below. Each accepted target therefore has a specified reference basis and an exact master expansion.

6.5 Source identities and the reference integral order

To extend the starting masters into targets, we first generated relations among the integrals of each family. We used the IBP identity in Eq. (7) with seeds in every nonempty physical sector within the bounds of Eq. (27). The auxiliary sunrise indices remained nonpositive. We fixed this candidate seed set before constructing targets, and retained each equation with its seed $\boldsymbol{\sigma}$, differentiated loop momentum ℓ_k and vector v [4, 2]. The resulting equations can contain integrals beyond the seed bounds because differentiation shifts the indices.

The vectors v included loop momenta, external momenta and propagator momenta q_i. Each propagator momentum is a fixed linear combination of loop and external momenta, so its IBP equation is the corresponding linear combination of the usual IBP equations at the same seed. We retained this decomposition to specify how each equation follows from its source operators. We also obtained symmetry relations by explicit changes of loop variables whose loop transformation matrix has determinant $+1$ or -1. We accepted a transformation when it permuted the inverse propagators with their masses unchanged and preserved the physical and auxiliary index positions.

To give the constructions a reduction order, we protected the chosen masters from elimination and ordered every other integral by

$$(S(\boldsymbol{a}),\, n_{\mathrm{dot}}(\boldsymbol{a}),\, s(\boldsymbol{a}),\, r(\boldsymbol{a}),\, \boldsymbol{a}). \tag{36}$$

Here S is the sector number in Eq. (5), and the remaining quantities are the index sums in Eq. (14). We compare the entries in the displayed order, using the first unequal entry to decide which integral is larger. We compare the components of $\boldsymbol{a}$ in the same way when all earlier entries agree. The leading integral of an equation is its largest integral outside the master basis. This reference order specifies the construction and recorded reductions; adapting the elimination model to Kira will require its actual ordering and available pivot choices.

6.6 Reverse construction from combinations of masters

To obtain target expressions with known answers, we selected subsets of masters from the same family and replaced their terms through the source identities. Let $\mathcal{J} = \{m_1, \ldots, m_h\}$ be the selected master labels, where h is the number of chosen masters. The initial expression is

$$\mathcal{T}_0 = \sum_{m \in \mathcal{J}} c_m(\boldsymbol{x}, d) M_m, \qquad 1 \le h \le \min(3, n_M). \tag{37}$$

The coefficients are rational functions of the independent kinematic parameters and $d = 4 - 2\epsilon$, with no dependence on the loop momenta. Writing them in terms of d is equivalent to writing them in terms of ϵ. Their mass dimensions must compensate for differences between the chosen masters so that every term in $\mathcal{T}_0$ has the same dimension.

For this purpose, let $B_m = \sum_i \mu_i^{(m)}$ be the sum of the indices of master M_m, and set $A = B_{m_1}$. For a fixed nonzero squared mass or momentum invariant Q^2 in the family, we used

$$c_{m_j} = u_j(d)(Q^2)^{B_{m_j}-A}, \qquad [\mathcal{T}_0]_{\text{mass}} = Ld - 2A. \tag{38}$$

Here $j = 1, \ldots, h$ identifies a selected master and u_j is dimensionless. We varied the weights through the three forms

$$\begin{gathered} u_j(d) \in \left\{ \eta_j,\, \eta_j \frac{d-2}{d-4},\, \eta_j \frac{j+1}{d-2} \right\}, \\ \eta_j \in \{-3, -2, -1, 1, 2, 3\}. \end{gathered} \tag{39}$$

Here η_j is a nonzero integer weight. For each sequence we chose one of these three forms and varied the integer weights and selected masters. The power of Q^2 in Eq. (38) gives c_m mass dimension $2(B_m - A)$, so Eq. (34) gives the stated common dimension.

To replace a term, write the current expression and a source IBP identity as

$$\mathcal{T}_n = \sum_{\boldsymbol{b}} w_{\boldsymbol{b}}^{(n)} I_{\boldsymbol{b}}, \qquad R = \sum_{\boldsymbol{b}} r_{\boldsymbol{b}} I_{\boldsymbol{b}} = 0. \tag{40}$$

Here n is the number of completed replacements, while $w_{\boldsymbol{b}}^{(n)}$ and $r_{\boldsymbol{b}}$ are the coefficients of integral $I_{\boldsymbol{b}}$ in the expression and identity. The sums contain their finitely many nonzero terms. For a present term $I_{\boldsymbol{a}}$, we selected an identity generated at seed $\boldsymbol{a}$ with $r_{\boldsymbol{a}} \neq 0$, and formed

$$\mathcal{T}_{n+1} = \mathcal{T}_n - \frac{w_{\boldsymbol{a}}^{(n)}}{r_{\boldsymbol{a}}} R. \tag{41}$$

This operation cancels the chosen term and introduces other family integrals. Since $R = 0$, the resulting expression has the same master expansion as $\mathcal{T}_0$. We required the identity to introduce a leading integral larger than every integral already present, so the recorded steps can be reversed through the order in Eq. (36).

We attempted 24 sequences per family, with at most four replacements per sequence, and retained accepted intermediate expressions. A sequence stopped when no admissible replacement was found within the construction limits. We removed identical target expressions and retained only nonzero targets with complete reference reductions. When the final expression contained one integral with coefficient $q \neq 0$, we divided the expression and its master coefficients by q and recorded this normalization. Expressions containing several index vectors remain targets for reduction of the complete sum. To obtain further individual integrals, we also applied the source identities in the following way.

6.7 Constructing individual integrals from known reductions

Starting from Eq. (35), we searched for a source identity whose leading integral was the only term without a known master expansion. Such an identity has the form

$$h_0 I_{\boldsymbol{b}} + \sum_{j=1}^{n_R} h_j I_{\boldsymbol{a}^{(j)}} = 0, \qquad h_0 \neq 0, \tag{42}$$

where $I_{\boldsymbol{b}}$ is the new integral, n_R is the number of other terms and h_0, h_j are the identity coefficients. Substituting the known reductions of the other terms gives

$$I_{\boldsymbol{b}} = \sum_{m=1}^{n_M} C_{\boldsymbol{b}m} M_m, \qquad C_{\boldsymbol{b}m} = -\sum_{j=1}^{n_R} \frac{h_j}{h_0} C_{\boldsymbol{a}^{(j)}m}. \tag{43}$$

We added each resulting expansion to the known set so later identities could use it. If all its master coefficients vanished, we retained the zero-integral proof as a dependency for subsequent constructions. The target collection contains the accepted nonzero examples.

We made at most five passes through the source equations and allowed at most 64 new nonzero constructions of this type per family. An equation with several still-unknown integrals became usable only after other identities supplied the missing expansions. Solving remaining coupled systems would extend this construction beyond the individual integrals obtained here. Both this procedure and the reverse replacements above preserve the source equations and their dependencies, which determine the seed and elimination records.

6.8 Seed records and elimination sequences

For each accepted target, we retained its exact master coefficients together with the source identities used to obtain them. An IBP identity contributes its seed and derivative operator, while a symmetry identity contributes its loop-variable transformation and integral mapping. Dependencies carry the earlier identities needed to reduce intermediate integrals or establish that they vanish. For an IBP step of the form in Eq. (42), the sufficient seed set therefore obeys

$$\mathcal{S}_{\boldsymbol{b}} = \{\boldsymbol{\sigma}\} \cup \bigcup_{j=1}^{n_R} \mathcal{S}_{\boldsymbol{a}^{(j)}}, \qquad \mathcal{S}_{\boldsymbol{\mu}^{(m)}} = \varnothing. \tag{44}$$

Here $\mathcal{S}_{\boldsymbol{a}}$ is the seed set recorded for $I_{\boldsymbol{a}}$, $\boldsymbol{\sigma}$ is the seed of the current IBP identity and $\varnothing$ is the empty set. Symmetry steps retain the dependency sets and their symmetry record. For reverse constructions, the seed set is the union of the seeds of the identities used along the recorded path.

We retained the full candidate seed set alongside each family so the seed model can compare possible choices. The recorded subset supplies enough equations for the known reduction, and seeds outside it can still contribute to other valid reductions. For the model inputs, we stored the family, basis, target coefficients and index information, including the sector and $t, r, s, n_{\mathrm{dot}}$. We kept the exact answer, construction history and reference seed choices separately.

To obtain elimination examples, we represented the target as a row of coefficients $T_{\boldsymbol{b}}$ multiplying $I_{\boldsymbol{b}}$. At each step, we selected its largest remaining integral outside the master basis, denoted by $I_{\boldsymbol{a}}$. The available pivot rows were the retained source identities with leading integral $I_{\boldsymbol{a}}$ and a nonzero coefficient for that integral. For the chosen row R, the update was

$$T_{\boldsymbol{b}} \;\longleftarrow\; T_{\boldsymbol{b}} - \frac{T_{\boldsymbol{a}}}{r_{\boldsymbol{a}}} r_{\boldsymbol{b}}. \tag{45}$$

This cancels $I_{\boldsymbol{a}}$ from the target and can introduce only smaller integrals or masters. Repeating the recorded choices gives the target's known master expansion. The source equations remain unchanged during this target-row reduction, so the current row, source equations and reference order specify the state seen by the elimination model.

For every step, we retained the row before and after the update, all available pivot rows, the selected equation and integral, the multiplier, and the numbers of new and cancelled nonzero entries. We kept the coefficients as exact rational functions and also stored numerical row states over a finite field. The first sample in Eq. (30), with $P_1 = 2147483647$ and $d_1 = 37$, supplied these numerical states. Substituting parameters into the coefficient functions leaves the integrals symbolic and uses exact modular arithmetic for the row operations [11].

For a selected source row with n_{row} nonzero terms, we counted one finite-field division and n_{row} multiplications and subtractions. Summing these counts gives the cost of the recorded target-row

sequence. Equation generation, forming combined IBP operators, symbolic simplification, model inference and reconstruction fall outside this count. The construction gives a valid reference path and an upper bound on the minimum cost under these rules. Establishing a minimum would require a separate argument covering alternative seed sets and elimination paths.

6.9 Completed collection and consistency checks

The collection contains 1,662 accepted targets across all 16 families, with 309 individual integrals, 1,353 integral expressions and 5,138 recorded elimination steps [13]. We divided the examples by complete families into 1,290 training targets, 185 validation targets and 187 test targets. Validation targets belong to the unequal-mass bubble and massless triangle with three off-shell legs. Test targets belong to the massless box with four off-shell legs and the equal-mass sunrise. These held-out targets contain terms in the full physical sector, while their basis definitions still include the required lower sectors.

The validation collection consists entirely of integral expressions, and the test collection contains one individual integral and 186 expressions. Tests focused on individual-integral generalization will therefore require further held-out individual constructions. The present collection covers the specified one-loop and two-loop families and the stated construction bounds. Training the models and measuring their performance on larger reductions remain later stages.

Before accepting the collection, we rederived all 55,654 stored source equations from their family definitions and checked the recorded symmetry transformations. We also checked 2,699 used IBP equations by independently differentiating the original quadratic expressions for the propagators in terms of loop scalar products. We then replayed every target construction and elimination sequence with exact rational functions, including intermediate zero-integral dependencies and final normalizations. Each final result agreed with its stored master expansion, and the checks also confirmed mass dimensions, seed records and allowed pivots.

To check the sampled coefficients, we repeated every elimination sequence at three finite-field points. The first two were the samples in Eqs. (30) and (32), with the primes and dimension values specified there. The third used $P_3 = 1000000009$, $d_3 = 43$ and

$$\boldsymbol{z}^{(3)} = (-59, -73, -41, -61, -83, -101, 17, 17, 47), \tag{46}$$

in the parameter order of Eq. (29). Each family used its own parameters and retained its prescribed zeros and equal masses. All required divisions were defined at the three points, and every sampled reduction reproduced the reference coefficients.

The exact symbolic checks establish the recorded reductions from their source identities at generic parameters, while the three-point checks verify the numerical data supplied with them. This construction already provides the symbolic coefficient functions, so obtaining the reference answers requires no rational-function reconstruction. The accompanying data contain the targets, equations, construction histories, sufficient seeds and elimination records needed to train and test the two proposed models [13].

Citations

[1] Stefan Weinzierl. Introduction to Feynman integrals, 2010. arXiv:1005.1855.

[2] Philipp Maierhöfer, Johann Usovitsch, and Peter Uwer. Kira—a Feynman integral reduction program. *Computer Physics Communications*, 230:99–112, 2018. arXiv:1705.05610.

[3] Stefan Weinzierl. *Feynman Integrals: A Comprehensive Treatment for Students and Researchers*. UNITEXT for Physics. Springer, 2022. arXiv:2201.03593; publisher DOI.

[4] K. G. Chetyrkin and F. V. Tkachov. Integration by parts: The algorithm to calculate β-functions in 4 loops. *Nuclear Physics B*, 192:159–204, 1981. publisher DOI.

[5] Thomas Gehrmann and Ettore Remiddi. Differential equations for two-loop four-point functions. *Nuclear Physics B*, 580:485–518, 2000. arXiv:hep-ph/9912329.

[6] A. V. Smirnov and A. V. Petukhov. The number of master integrals is finite, 2010. arXiv:1004.4199.

[7] Stefano Laporta. High-precision calculation of multiloop Feynman integrals by difference equations. *International Journal of Modern Physics A*, 15:5087–5159, 2000. arXiv:hep-ph/0102033.

[8] Philipp Maierhöfer and Johann Usovitsch. Kira 1.2 release notes, 2018. arXiv:1812.01491.

[9] Fabian Lange, Johann Usovitsch, and Zihao Wu. Kira 3: Integral reduction with efficient seeding and optimized equation selection, 2025. arXiv:2505.20197.

[10] Philipp Kant. Finding linear dependencies in integration-by-parts equations: A Monte Carlo approach. *Computer Physics Communications*, 185:1473–1476, 2014. arXiv:1309.7287.

[11] Jonas Klappert, Fabian Lange, Philipp Maierhöfer, and Johann Usovitsch. Integral reduction with Kira 2.0 and finite field methods. *Computer Physics Communications*, 266:108024, 2021. arXiv:2008.06494.

[12] Rushil Saraswat. Master-integral bases for sixteen one-loop and two-loop families. Accompanying family definitions, master bases and validation data, 2026.

[13] Rushil Saraswat. Target integrals constructed from master bases with seed and elimination records. Accompanying target integrals, source identities and validation data, 2026.